AuthService – Architecture & Topology Diagrams

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High-Level Architecture (Mermaid)

```
%% AuthService High-Level Architecture
flowchart LR
  Client[[Client / Swagger / Postman]]
  API[AuthService.Api (Controllers)]
  App[Application (MediatR CQRS, Validators)]
  Abs[Abstractions (Interfaces)]
  Dom[Domain (Entities, Value Objects)]
  Infra[Infrastructure (EF/Identity/JWT/Repos)]
  Write[(SQL Server - WriteDbContext)]
  Read[(PostgreSQL - ReadDbContext)]
  Aspire[(Aspire AppHost - optional)]

  Client --> API
  API --> App
  App --> Abs
  App --> Dom
  App --> Infra
  Infra --> Dom
  Infra --> Write
  Infra --> Read
  Aspire -. optional orchestration .-> API
  Aspire -. optional orchestration .-> Write
  Aspire -. optional orchestration .-> Read
```

Authentication Flows (Mermaid)

```
%% Authentication Flows
sequenceDiagram
  participant U as User
  participant API as API
  participant App as Application (MediatR)
  participant Id as Identity (UserManager/SignInManager)
  participant JWT as JwtTokenService
  participant SQL as WriteDbContext
  participant PG as ReadDbContext

  U->>API: POST /api/auth/register
  API->>App: RegisterCommand
  App->>Id: Create user + add role
  Id-->>SQL: Save user
  App-->>U: Confirmation link (log)

  U->>API: GET /api/auth/register/confirm?userId&token
  API->>App: ConfirmCommand
  App->>Id: Confirm Email
  App-->>U: OK

  U->>API: POST /api/auth/login
  API->>App: LoginCommand
  App->>Id: PasswordSignInAsync
  App->>JWT: CreateAccessToken + RefreshToken
  App-->>U: tokens

  U->>API: GET /api/users/me (Bearer)
  API->>App: GetProfileQuery
  App->>Id: Load user + roles
  App-->>U: Profile

  U->>API: POST /api/users/addresses (Bearer)
  API->>App: CreateAddressCommand
  App->>SQL: Insert Address
```

```
  App-->>U: Created
  U->>API: GET /api/users/addresses (Bearer)
  API->>App: GetAddressesQuery
  App->>PG: Query Addresses
  App-->>U: List
```

Compose + Aspire Topology (Mermaid)

```
%% Docker Compose + Aspire Resource Topology
graph TD
  subgraph Compose
    A[auth_api] -->|depends_on| B[(sqlserver)]
    A -->|depends_on| C[(postgres)]
  end

  subgraph Aspire (optional)
    H[AppHost]
    H --> A
    H --> B
    H --> C
  end

  A -->|:8080| User[Client]
  B -->|:1433| DBA1[SSMS]
  C -->|:5432| DBA2[psql]
```